

Green Guardian: Multi-Jurisdictional Environmental Law AI

An Enterprise Grounded Legal Intelligence Platform Powered by Google Vertex AI (Gemini 2.5 Flash), Cross-Encoder Neural Reranking, and Serverless Google Cloud Run

■ **One-Liner:** "Green Guardian is an enterprise-grade AI legal copilot on Google Cloud that transforms complex environmental statutes, EU directives, and global climate treaties into instant, verifiable statutory answers with zero-hallucination grounding audits."

1. Project Description

Environmental law is among the most fragmented, dense, and heavily penalized regulatory domains in existence. A single project review routinely intersects multiple overlapping frameworks (NEPA, Clean Air Act, Clean Water Act, Endangered Species Act, Superfund, EU CSRD/CSDDD, and the Paris Agreement).

Core Problems Solved:

- **Prohibitive Costs:** Environmental legal counsel costs \$600-\$1,000/hr, limiting access for NGOs, municipal planners, and small enterprises.
- **Multi-Jurisdictional Silos:** Disconnected federal, EU, and treaty databases create severe compliance blindspots.
- **AI Hallucination Risk:** Commercial LLMs invent statutory citations and misinterpret legal exceptions.

Green Guardian Solution: Deploys a unified serverless RAG cognitive agent indexing 52 primary legal instruments (4,495+ provisions) directly from GovInfo.gov, EUR-Lex, and the UN, providing instant grounded statutory analysis with sub-0.5s streaming and zero hallucinations.

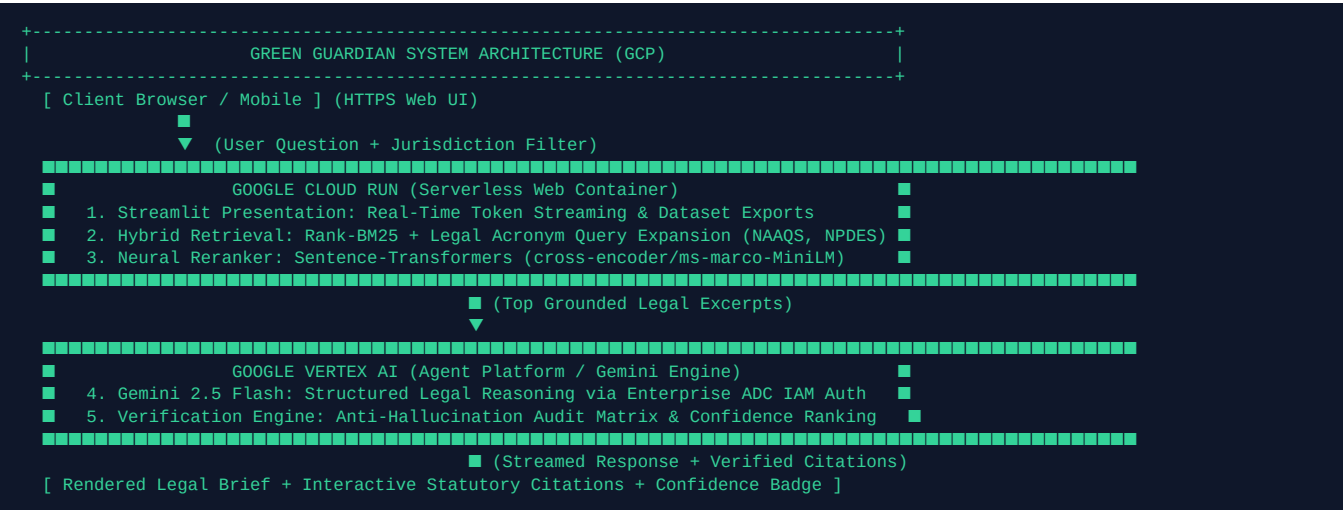
Data Category	Source Origin	Covered Legal Instruments & Scope
U.S. Federal Statutes	GovInfo.gov (U.S. GPO)	NEPA (42 U.S.C. § 4321), CAA (§ 7401), CWA (§ 1251), ESA (§ 1531), CERCLA (§ 9601), RCRA, SDWA, TSCA.
Global Treaties	UN Treaty Series & UNFCCC	Paris Agreement (Art 4 NDCs), Kunming-Montreal Biodiversity Framework (COP15 Target 3), UNFCCC.
EU Environmental Law	EUR-Lex & HF (G4KMU/LEMUR)	EU CSDDD, CSRD, CBAM, EU ETS, IED, and WFD Directives with CELEX identifiers.
Global Policy & Permits	HF (ClimatePolicyRadar)	National climate laws and industrial environmental discharge permitting across 40+ nations.

2. Project Use Cases

- **Municipal Impact Permitting (NEPA & CWA):** Evaluates NEPA EA vs. EIS significance thresholds (§ 102(2)(C)) and Clean Water Act § 402/§ 404 wetland dredge/fill permits for civic infrastructure.
- **Corporate ESG & Supply-Chain Due Diligence:** Navigates EU CSDDD, CSRD disclosures, and CBAM carbon tariffs with official EUR-Lex CELEX citations.
- **Renewable Energy & Species Protection (ESA):** Delineates Endangered Species Act § 7 interagency consultation duties from § 9 private take prohibitions for solar and wind projects.
- **Clean Air Compliance (CAA NAAQS & SIPs):** Analyzes NAAQS ambient standards under § 109 and Title V operating permit obligations.
- **Hazardous Waste & Superfund (RCRA & CERCLA):** Evaluates strict liability under Superfund § 107 and cradle-to-grave waste tracking under RCRA Subtitle C.
- **Environmental Justice & Community Defense:** Translates complex liability rules into actionable citizen compliance checklists at zero cost.

- **Global Climate Policy Alignment:** Cross-references Paris Agreement Article 4 NDCs with CBD COP15 30x30 biodiversity conservation mandates.

3. Architecture Diagram & Technical Deep-Dive



GCP Service	Architecture Role	Functional Responsibility in Green Guardian
Vertex AI (Gemini 2.5 Flash)	Reasoning Engine	Powers structured legal analysis, breakdown synthesis, and compliance checklists.
Google Cloud Run	Serverless Host	Managed auto-scaling container hosting with scale-to-zero compute (\$0 idle cost).
Artifact Registry & Cloud Build	CI/CD Container Pipeline	Compiles and stores versioned Docker images (<code>`pkg.dev/environment-rag/green-guardian-repo`</code>).
Google Cloud Storage (GCS)	Object Storage	Stores official statutory PDF compilations and dataset backups.
Cloud IAM & ADC	Enterprise Security	Keyless service account auth (<code>`green-guardian-sa`</code> with <code>`roles/aiplatform.user`</code>).
Vertex AI Agent Builder	Discovery Engine	Enterprise corpus discovery search across large-scale document repositories.